End-Term Sample Question Paper - SET D

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set D - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State the two steps of a proof by mathematical induction, and explain in one sentence why a proof that omits the basis step is worthless. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Q2. Given f(0,0) = 10, f(1,0) = 30, f(0,1) = 50 and f(1,1) = 70, compute the intensity at (0.25, 0.75) by bilinear interpolation, and state the nearest-neighbour value. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

Q3. Given P(A) = 0.7, P(B) = 0.5 and P(A ∩ B) = 0.35, compute P(A | B) and state whether A and B are independent, with the reason. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Q4. State the condition a directed graph must satisfy for a topological ordering to exist, and give a one-line reason. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Q5. State the worst-case running time of merge sort and of quick sort, and state the condition that triggers quick sort's worst case. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. Six activities are available with (start, finish) times (1,4), (3,5), (0,6), (5,7), (8,11) and (8,12). Apply the greedy activity-selection algorithm: state the sorting key, list the activities chosen, and state how many are selected. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Q7. Simplify F(A, B, C) = Σm(1, 3, 5, 7) using a three-variable Karnaugh map. Write the minterms in binary, state the single group that covers them, give the minimal sum-of-products expression, and state the literal saving. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Q8. Explain what the correlation coefficient measures and state its range; state what R^2 represents; and explain in two sentences why a strong correlation does not establish causation. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

Q9. State the Master theorem's three cases, and give one recurrence to which it does **not** apply, explaining why. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Linear Discriminant Analysis. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) State what Fisher's criterion maximises, and explain why LDA uses the within-class scatter rather than simply joining the two class means. *[5]*

(b) For $C_1 = \{(1,2), (3,2), (2,5)\}$ and $C_2 = \{(6,4), (8,4), (7,7)\}$: compute both means, S_1, S_2 and S_W, the direction $w = S_W^{-1}(m_2 - m_1)$, and the threshold midway between the projected means. *[5]*

Q11. Network flow. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) State the max-flow min-cut theorem, explain what a residual graph is, and describe how the minimum cut is read off once the algorithm stops. *[5]*

(b) Find the maximum flow for capacities s→a = 12, s→b = 6, a→b = 3, a→t = 9, b→t = 8. Give each augmenting path and bottleneck, the maximum flow, the reachable set S, and the cut capacity. *[5]*

Q12. Genetic algorithms. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) Describe the encoding, fitness, selection, crossover and mutation operators, and explain the role of elitism. *[5]*

(b) A GA maximises $f(x) = x^2$ over 5-bit chromosomes starting from 01011, 11100, 00110, 10001. Decode each, compute the fitness, the total and average, and the roulette-wheel selection probabilities. *[5]*

----- End of Question Paper -----